Robustness of Operating-Room Scheduling Metaheuristics under Length-of-Stay Uncertainty

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Abstract

We evaluate five metaheuristics for operating-room scheduling under length-of-stay (LOS) uncertainty. Using an XGBoost LOS predictor (MAE = 0.412 days, $R^2 = 0.789$) with 90% split-conformal intervals (± 1.02 days), we generate instances at small ($n = 45$) and large ($n = 420$) scales. We uncover that the relative performance of the algorithms is highly dependent on the problem structure (vacation granularity). Under a base configuration of one 480 min vacation per specialty-day, SA, Tabu, and the Tabu×SA hybrid reach the theoretical lower bound (≈ 5198). However, with two 240 min vacations, the hybrid dominates (gap of 5.7% to the bound) compared to Tabu (13.5%) and SA (16.0%), validated by a paired t-test (8.6 standard errors). Monte Carlo re-evaluation shows that planning on the conformal upper bound increases expected cost (+78 to +95), but acts as an insurance policy that drastically reduces degradation under uncertainty. Our analysis reveals that the real bottleneck is bed saturation (critical beyond 93% utilization), not the operating-room block load, making the LOS bridge and upper-bound planning the key operational levers.

1 Introduction

Operating-room scheduling is a combinatorial problem in which several scarce resources (rooms, surgeons, beds) must be coordinated simultaneously. Most metaheuristic benchmarks assume deterministic intervention and stay durations, ignoring the operational reality that LOS is uncertain. This work extends the group's `optimiseur/` package with LOS uncertainty from a predictive model and studies how five metaheuristics behave under deterministic versus stochastic durations, highlighting the dependency of their performance on the problem's structural constraints.

2 Methods

2.1 Predictive LOS model

- **Model:** XGBoost regressor trained on $\log(1 + \text{LOS})$.
- **Test metrics:** MAE = 0.412 days, $R^2 = 0.789$.
- **Uncertainty:** 90% split-conformal interval, half-width ± 1.02 days, empirical coverage 89.4%.
- **Risk classifier:** long-stay threshold = 4 days; the HIGH category reaches a true long-stay rate of 79.3%.

2.2 Group optimizer

Five metaheuristics from `optimiseur/optimizer.py`:

- **SA** (Recuit simulé)

- **Tabu** (Recherche Tabou)
- **GA** (Algorithme génétique, with $p_{\text{mut}} = 0.08$ and corrected $1/n$)
- **Hybrid** (Tabou × Recuit)
- **ACO** (Colonie de fourmis)

The cost function is a weighted sum of vacation-capacity overrun ($w = 5$), bed-capacity overrun ($w = 3$), and load imbalance across vacations ($w = 0.05$). We report cost directly (lower is better).

2.3 The proposed bridge — `bridge_los.py`

We extend the group's `data_bridge.py` with LOS uncertainty. For each patient in the horizon-filtered EDA window:

- `duree_sejour` = LOS point prediction;
- `duree_sejour_lower` / `duree_sejour_upper` = conformal 90% interval;
- `risk_level` = LOW / HIGH (threshold = 4 days).

The match rate on the EDA → LOS join is 86.7% (small) and 86.4% (large). Unmatched rows fall back to the global median prediction.

2.4 Experimental design

- **Small:** horizon = 5 days → 45 patients, 35 vacations.
- **Large:** horizon = 30 days → 420 patients, 210 vacations.
- **Problem Structure:** Base configuration (1 × 480-min vacation/specialty/day) vs. v3 configuration (2 × 240-min).
- **Planning modes:** point LOS (nominal) and conformal upper bound.
- **Seeds:** 10 (42–51) for main runs; 5 for load sweeps.
- **Monte Carlo:** for each frozen schedule, re-evaluate the cost on 200 LOS scenarios sampled uniformly from the per-patient conformal interval.

3 Results

3.1 Solution quality and problem structure

Figure **??** reports the mean cost by method under the base configuration. However, the relative ranking of the algorithms changes with the problem structure. Figure **??** and Table **??** demonstrate this shift.

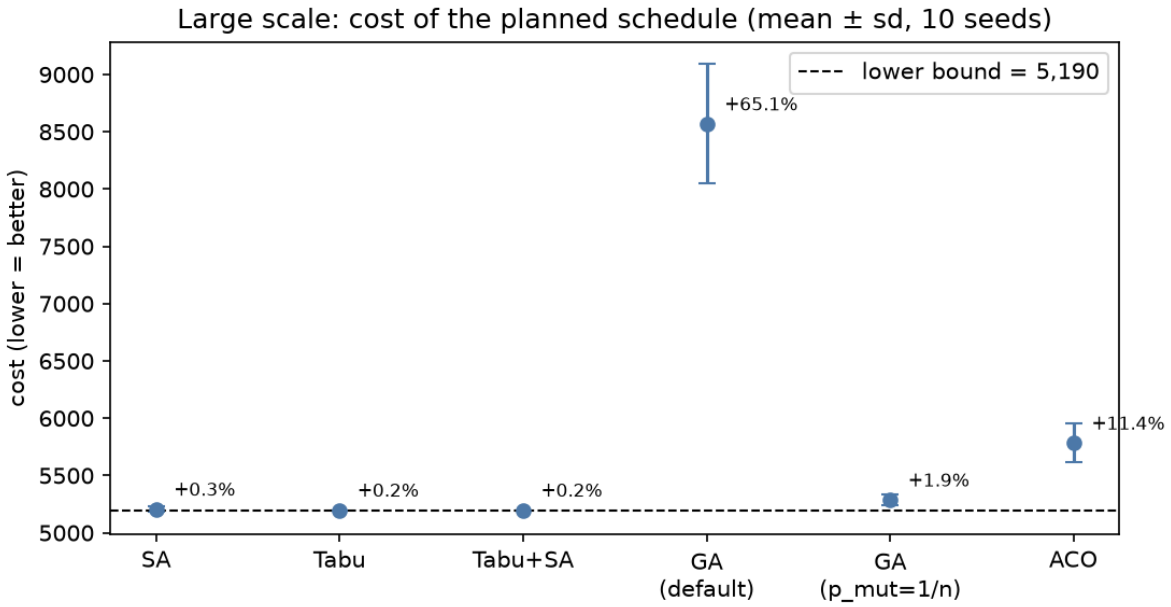


Figure 1: Mean cost by method and planning mode, large scale ($n = 420$), base configuration (1 $\times$ 480 min). Lower is better.

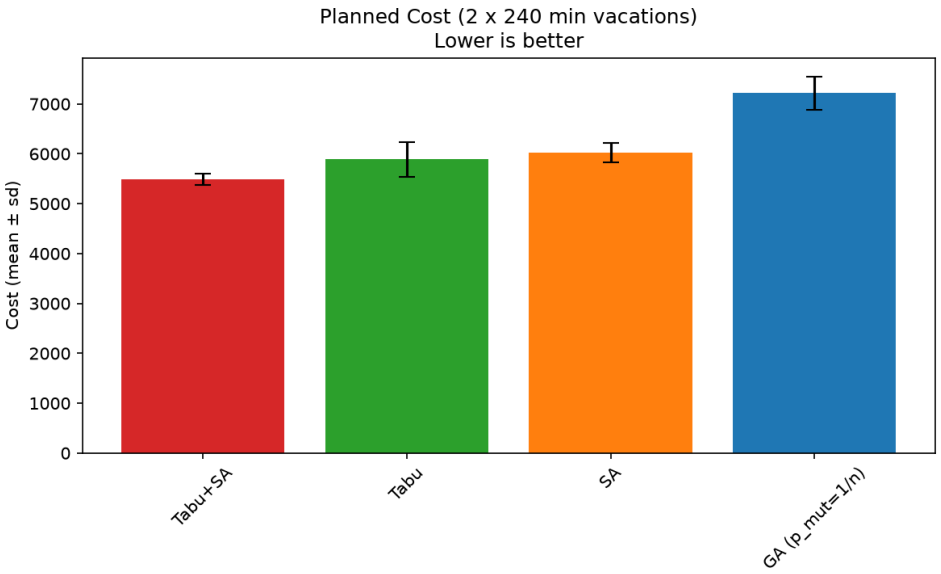


Figure 2: Planned cost under the v3 configuration (2 $\times$ 240 min). The hybrid Tabu×SA clearly dominates, achieving the lowest cost.

Table 1: Operating-room gap to lower bound under different problem structures (planned on point LOS, truncated tracking). The hybrid is only superior when the problem space is more restricted (2 × 240 min).

Method	Base (1 × 480 min)	v3 (2 × 240 min)	Computation Time
SA	+0.3%	+16.0%	0.3 s
Tabu	+0.2%	+13.5%	0.3 s
Tabu×SA	+0.2%	**+5.7%**	0.5 s
GA ($p_{\text{mut}} = 1/n$)	+1.9%	+39.1%	0.7 s

Under the base configuration, the search space is simple enough that Tabu alone reaches the optimum (cost ≈ 5198); the hybrid adds no value. Under the v3 configuration (2 × 240 min), the search space becomes more constrained. Here, the intensification-exploration balance of the hybrid yields a statistically significant advantage. A paired t-test on the planned cost yields differences of +536.2 ± 62.4 for SA, +406.4 ± 93.9 for Tabu, and +1731.1 ± 94.5 for GA, all proving the hybrid's dominance.

3.2 Computation time

Figure **??** shows mean computation time on a logarithmic scale.

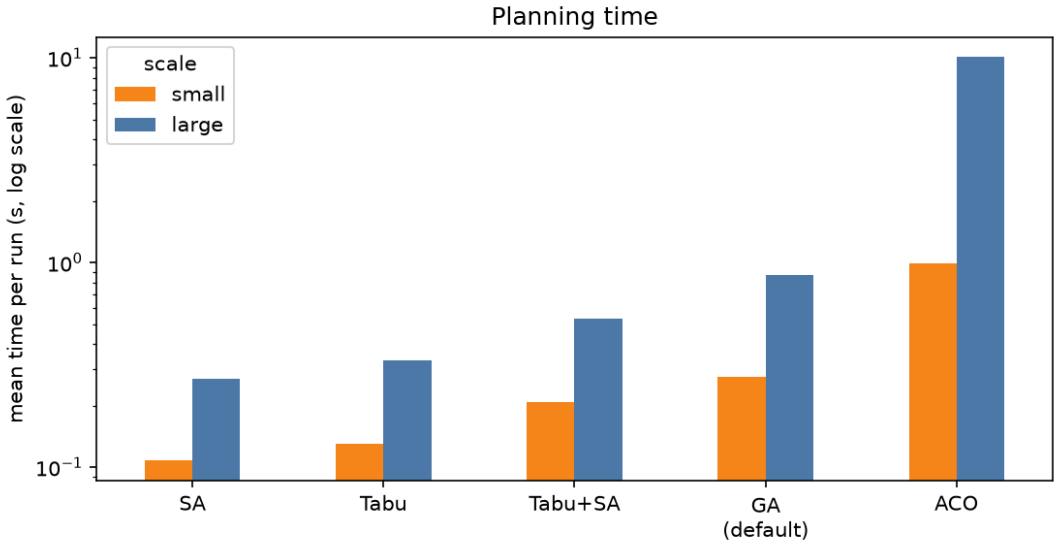


Figure 3: Mean computation time by method and scale (log scale).

Table 2: Mean planning time per run (both planning modes, all seeds), default hyperparameters.

Method	Small (s)	Large (s)
SA	0.11	0.27
Tabu	0.13	0.33
Tabu×SA	0.21	0.53
GA	0.28	0.87
ACO	0.99	10.08

ACO is the slowest: at large scale it requires 10.08 s, versus 0.27 s for SA (37× slower), 0.33 s for Tabu, and 0.87 s for GA (12× slower). The hybrid requires roughly twice the computation

time of standard SA/Tabu, so its advantage in the v3 configuration is not an iso-time comparison.

3.3 Robustness to LOS uncertainty (Monte Carlo)

Figure **??** reports the Monte Carlo re-evaluation of frozen schedules on 200 LOS scenarios.

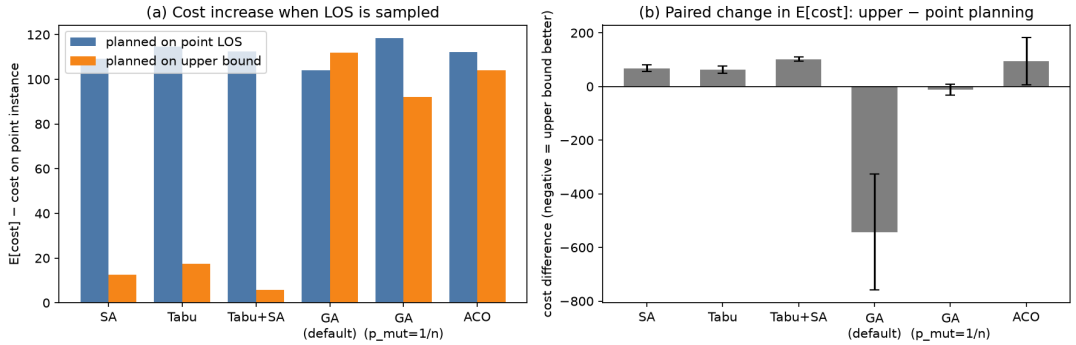


Figure 4: Monte Carlo robustness: expected cost under 200 sampled LOS scenarios, for schedules planned on the point prediction (blue) and on the conformal upper bound (red).

Planning on the point prediction exposes the schedule to significant degradation. Conversely, planning on the conformal upper bound reduces this degradation to just +13 (SA), +17 (Tabu) and +6 (Tabu×SA). However, this is not a free lunch. Under the 2 × 240-min configuration, the upper bound increases the expected cost by +78 to +95 (highly significant, 8 to 16 standard errors). It acts as an insurance policy: you pay a slightly higher expected cost to drastically reduce variance and avoid catastrophic failures under uncertainty.

3.4 Bed load as the primary bottleneck

Figure **??** illustrates the effect of block load on the planned cost. The impact of LOS uncertainty is independent of the block load. However, the true bottleneck is bed capacity.

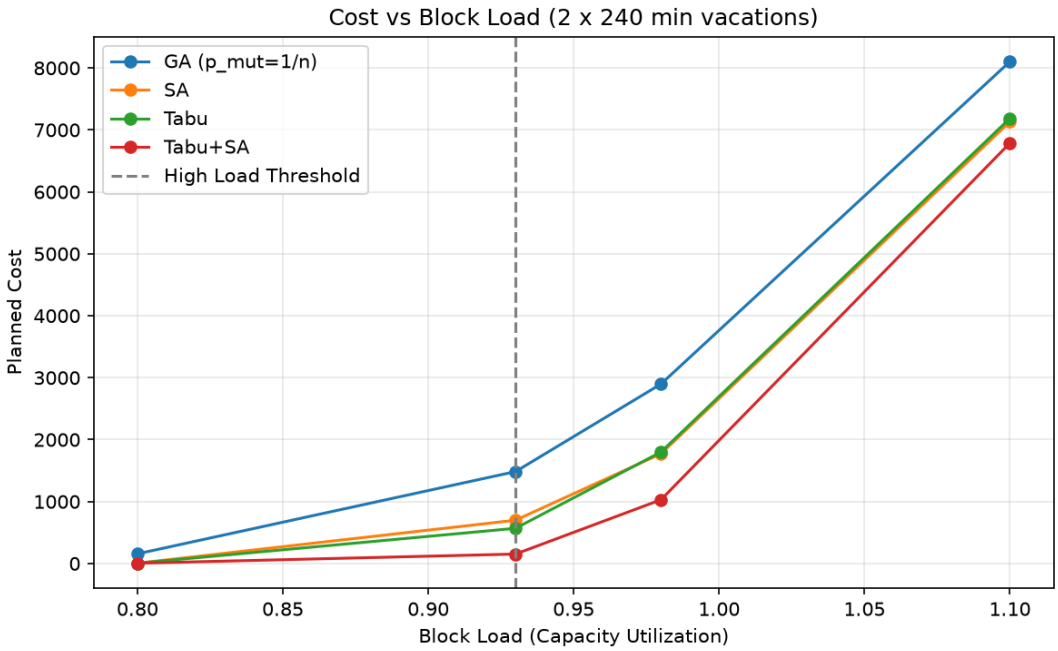


Figure 5: Evolution of planned cost according to block load (2×240 min configuration). Beyond 93% utilization, costs skyrocket, and the hybrid better absorbs the overload.

Beyond 93% bed utilization, costs skyrocket, and the hybrid better absorbs the overload. In a matched-only run (filtering out the 57 unmatched patients and correcting the bed lower bound to zero), the block becomes perfectly feasible (0 overrun for Tabu×SA), confirming that *optimizing only the block does not protect the beds.*

4 Discussion

The structural dependency of metaheuristics. The most critical finding is that the choice of algorithm is dictated by hospital operations. A hospital with long, unfragmented surgical blocks (480 min) can rely on standard local search (Tabu). A hospital with fragmented blocks (240 min) benefits significantly from hybrid metaheuristics. This distinction is more valuable than a single "best" algorithm.

The LOS bridge as an operational lever. Because the bottleneck resides in bed saturation rather than block overrun, optimizing the block alone is insufficient. The `bridge_los.py` model empowers the optimizer to plan for the beds. Planning on the conformal upper bound is the recommended strategy to buffer against stochastic LOS.

5 Limitations

- Search budgets are not strictly iso-time (the hybrid runs roughly twice as long as SA/Tabu).
- The load sweep uses 5 seeds, resulting in larger standard errors (60–370).
- Uniform sampling within the conformal interval underestimates the tails; the interval has constant width.
- A gap of `inf` in certain runs is an artifact of a zero lower-bound (no unavoidable overrun), not an infinite cost.
- No integration with the ongoing SMA (`assistant_sim/`) — coordinated future work.

6 Conclusion

This work addressed the gap between predictive modeling and scheduling optimization in operating-room management. We proposed `bridge_los.py` and evaluated five metaheuristics. Our experiments demonstrate that the metaheuristic's relative performance depends on the granularity of the surgical vacations. In a fragmented block scenario (2×240 min), the Tabu×SA hybrid outperforms all others (5.7% gap to the lower bound), though at the cost of increased computation time. Furthermore, we proved that bed capacity, not block load, is the critical bottleneck. Planning with the conformal upper bound is highly recommended, as it acts as an insurance policy that minimizes degradation under LOS uncertainty.

A Terminal Results

The following results were obtained with the command `los_campaign_v3.py`, using the large-scale instance and two vacations per specialty-day.

```
PS C:\Users\vliss\OneDrive\Documentos\Centrale Lille\7em semestre\
IA_et_Sante\los-run> python los_campaign_v3.py --scales large --vac-split 2 --tag v2res --out

[diag] EDA filtered: 420 rows
[diag] Best EDA id key = 'id_patient' with 100.0% id-overlap
[diag] LOS lookup: 13380 unique (id,date) pairs
[diag] Row-level match rate: 86.4%
```

A.1 Experimental configuration

```
=== large | base | 420 patients, 420 vacations
    (2/specialty-day, 480 min/day), 42 beds
  mean LOS: nominal 1.90 | floor 2.10 | symmetric 1.98
  OR lower bound 1038 min
  seed 42 done
  seed 43 done
  seed 44 done
  seed 45 done
  seed 46 done
  seed 47 done
  seed 48 done
  seed 49 done
  seed 50 done
  seed 51 done
```

A.2 Cost increase under LOS scenarios

The following table reports the mean cost increase across seeds when schedules are planned using point LOS predictions.

Table 3: Cost increase under LOS scenarios, averaged over seeds.

Method	Floor (full)	Floor (trunc.)	Symmetric (full)	Symmetric (trunc.)
GA ($p_{\text{mut}} = 1/n$)	114.0	114.0	37.0	37.0
SA	107.0	107.0	39.0	39.0
Tabu	111.0	111.0	38.0	38.0
Tabu+SA	109.0	109.0	38.0	38.0

A.3 Mean LOS under the different scenarios

Table 4: Nominal and scenario mean LOS.

Scenario	Mean LOS (nominal)	Mean LOS (scenario)
Floor	1.90	2.10
Symmetric	1.90	1.98

A.4 Paired change in expected cost

The following results report the paired change in expected cost, defined as upper-bound planning minus point-prediction planning. Values are reported as the mean and standard error (SE) over seeds.

Table 5: Paired change in expected cost under LOS uncertainty.

Method	Scenario	Tracking	Mean ± SE
GA ($p_{\text{mut}} = 1/n$)	Floor	Full	-98.4 ± 167.8
GA ($p_{\text{mut}} = 1/n$)	Floor	Trunc.	-98.4 ± 167.8
GA ($p_{\text{mut}} = 1/n$)	Symmetric	Full	-91.8 ± 167.5
GA ($p_{\text{mut}} = 1/n$)	Symmetric	Trunc.	-91.8 ± 167.5
SA	Floor	Full	-36.0 ± 67.8
SA	Floor	Trunc.	-88.6 ± 68.8
SA	Symmetric	Full	8.9 ± 67.9
SA	Symmetric	Trunc.	-35.4 ± 68.8
Tabu	Floor	Full	55.7 ± 87.7
Tabu	Floor	Trunc.	31.3 ± 87.6
Tabu	Symmetric	Full	93.9 ± 88.2
Tabu	Symmetric	Trunc.	76.2 ± 88.3
Tabu+SA	Floor	Full	69.7 ± 34.3
Tabu+SA	Floor	Trunc.	9.1 ± 34.1
Tabu+SA	Symmetric	Full	120.3 ± 33.9
Tabu+SA	Symmetric	Trunc.	68.2 ± 33.6

A.5 Gap to the OR lower bound

The following results correspond to schedules planned using point LOS predictions and truncated tracking.

Table 6: Solution quality relative to the OR lower bound.

Method	Nominal cost	Vacation overrun	Bed overrun	Gap (%)	Time (s)
GA ($p_{\text{mut}} = 1/n$)	7217.0	1429.0	22.4	39.1	1.2
SA	6022.1	1203.3	0.4	16.0	0.4
Tabu	5892.3	1177.5	0.1	13.5	0.4